

Improving Loan Default Prediction with Stacked Ensemble Models and Domain-Informed Feature Engineering

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1. Abstract

Accurate and responsible credit risk assessment is critical in the financial sector, particularly in determining loan eligibility and minimizing default rates. This study examines loan default prediction using the Home Credit Default Risk dataset from Kaggle. Multiple machine learning models and data preprocessing strategies were evaluated, with the best performance achieved through a stacked model composed of LightGBM, Random Forest Regressor, and Elastic Net Regressor as base learners, and Logistic Regression as the meta-learner. The results also indicate that class imbalance handling through undersampling and feature selection based on LightGBM feature importance were the most effective approaches on this dataset. These findings highlight the importance of ensemble learning and domain-informed feature engineering in improving credit risk prediction performance on large, noisy, and imbalanced financial datasets.

2. Introduction

Loan default is a significant challenge for credit institutions as it can lead to substantial financial losses if not prevented. To mitigate this risk, lenders conduct thorough assessments to estimate the likelihood of default and make informed decisions about whether to extend credit, often avoiding loans to individuals deemed high-risk. Developing a loan default model involves tackling three key challenges: ensuring predictive accuracy, maintaining fairness across protected groups, and providing explainability to support decision-making and recourse.

The Home Credit Default Risk dataset, released on Kaggle as part of a data science competition, has served as a benchmark for evaluating loan default models. It is notable for its rich, relational structure, incorporating multiple tables with historical loan, credit card, and installment payment data. Researchers have applied a variety of ML techniques to this dataset, including feature engineering, dimensionality reduction, and boosting algorithms, to maximize predictive accuracy.

The dataset includes demographic, financial, and behavioral information on applicants. The core `application_train` table contains the primary target variable `TARGET` (1 if the loan is in default), and several supplementary tables provide historical data on credit cards, installment payments, bureau credit history, and more. This multi-source format enables the creation of enriched features for predictive modeling.

Traditional credit scoring models, such as logistic regression, have long been favored for their interpretability and regulatory acceptance. While these models remain valuable, their linear assumptions can limit performance when applied to complex borrower behavior and nonlinear feature interactions. In response, ensemble-based machine learning models including tree-based methods and model stacking, have demonstrated superior predictive capability in many credit risk applications.

Tree-based ensembles like Random Forests, XGBoost, and LightGBM are widely used in credit scoring for their high accuracy and ability to handle class imbalance. Akinjole et al. (2024) report $\approx 93.7\%$ accuracy using a stacked model with XGBoost, AdaBoost, and deep learning; Zhu et al. (2024) also utilized LightGBM-XGBoost ensembles.

Nevertheless, improvements in model performance are not solely driven by algorithmic complexity. Effective preprocessing, feature engineering, and the incorporation of domain knowledge are often decisive factors, particularly when working with heterogeneous and imperfect financial data.

In particular, this work emphasizes the role of domain-informed decision-making in the modeling pipeline. Techniques such as class imbalance handling through undersampling and feature selection guided by LightGBM feature importance were systematically evaluated. The results demonstrate that these strategies, when informed by both statistical evidence and domain understanding, contribute substantially to improved predictive performance. By highlighting the interplay between ensemble learning, targeted preprocessing, and domain knowledge, this study aims to provide practical insights for developing robust credit risk models in real-world financial settings.

3. Data Preprocessing and Feature Engineering

3.1 Table Aggregation

The Home Credit Default Risk dataset consists of a primary application-level table accompanied by multiple auxiliary tables capturing borrowers' historical and behavioral information. To construct a unified modeling dataset, all relevant auxiliary tables including *installments_payments*, *bureau*, and *previous_application* were aggregated and joined to the *application_train* table using *SK_ID_CURR* as the common identifier. For each auxiliary table, statistical aggregation functions such as mean, sum, count, and frequency-based summaries were computed at the applicant level.

3.2 Feature Engineering

A total of **713 new features** were generated through a combination of manual feature engineering, table aggregation, label encoding, one-hot encoding, and currency-based pivoting. Feature design was guided

by both empirical performance considerations and domain knowledge in credit risk assessment, with particular emphasis placed on repayment behavior and borrowing patterns.

Installment payment behavior was extensively engineered, as repayment history is a strong indicator of creditworthiness. The following are some of the domain-relevant features that were derived from the *installments_payments* table:

- **PAID_COMPLETE**: Total monetary value of fully paid installments.
- **PAID_ONTIME**: Number of installments paid on or before the contractual due date.
- **PAID_COMPLETE_WEIGHTED**: Time-weighted sum of fully paid installments, assigning higher weights to more recent payments to capture recency effects.
- **PAID_ONTIME_WEIGHTED**: Time-weighted count of on-time payments, emphasizing recent repayment behavior.

In addition, features derived from the *previous_application* table were designed to reflect borrowing intent and credit allocation outcomes. Notably, this feature was introduced:

- **APP_CREDIT_RATIO**: Ratio of the amount requested (*AMT_APPLICATION*) to the amount granted (*AMT_CREDIT*), capturing borrower optimism or overconfidence in credit demand.

Lastly, features derived from the bureau table were engineered to capture borrowers' external credit exposure, repayment progress, and utilization behavior across active and closed credit accounts. Since bureau data reflect credit relationships beyond the current lender, these features provide critical signals regarding a borrower's overall debt burden and credit management patterns.

Several features were designed to incorporate temporal dynamics by weighting credit-related quantities using **DAYS_CREDIT_UPDATE**, thereby emphasizing more recently updated credit information. These are some of the bureau-based features that were introduced:

- **REMAINING_DEBT_PER_DAY**: Estimates the borrower's remaining debt adjusted by the expected credit end date and recency of bureau updates, providing a time-sensitive measure of outstanding obligations.
- **CREDIT_PROGRESS**: Captures the relative progress of a credit account toward its scheduled end date, weighted by recency. This feature reflects whether a borrower is progressing as expected through a credit contract.
- **CREDIT_PER_DAY**: Measures average outstanding debt per remaining credit duration, adjusted for recency, representing the intensity of ongoing credit exposure.
- **CREDIT_UTILIZATION**: Ratio of total credit amount to the approved credit limit, weighted by recency, indicating how aggressively available credit is being used.
- **DEBT_UTILIZATION**: Ratio of outstanding debt to credit limit, weighted by recency, reflecting the borrower's current leverage level.
- **PAID_AMOUNT**: Total amount already repaid on a credit account, adjusted by recency, serving as an indicator of repayment capability and progress.
- **OVERDUE_WEIGHTED**: Recency-weighted measure of overdue credit amounts, emphasizing recent delinquency behavior.

- **CREDIT_ACTIVE_WEIGHTED_RECENCY**: Recency-weighted indicator of active credit accounts, capturing ongoing external credit exposure.
- **TOTAL_CLOSED**: Aggregated credit amount of closed accounts, weighted by recency, reflecting completed credit obligations.
- **TOTAL_DEBT**: Recency-weighted sum of outstanding debt from active credit accounts.
- **TOTAL_DEBT_OVERDUE**: Recency-weighted sum of overdue amounts from active credit accounts, directly capturing recent default risk signals.

These engineered features aim to encode behavioral and contextual information that may not be directly observable from raw financial variables alone.

3.3 Evaluation of Alternative Preprocessing Strategies

In addition to the preprocessing techniques retained in the final pipeline, several alternative preprocessing and data-handling strategies were explored and embedded directly within the modeling process. These techniques were evaluated empirically to assess their impact on predictive performance before being included in the final model configuration.

Among the techniques evaluated, **random undersampling of the majority class** consistently improved model performance by mitigating the effects of severe class imbalance, leading to better discrimination of default cases. As a result, this approach was incorporated into the final modeling pipeline.

Similarly, **feature selection based on LightGBM feature importance** proved effective in reducing dimensionality and eliminating noisy or weakly informative features. Other preprocessing strategies were tested but ultimately excluded from the final model due to limited or inconsistent performance gains.

3.4 Dataset Split

To evaluate model performance under realistic conditions, the dataset was divided into training and test sets using an 80/20 split, with 20% of the data reserved exclusively for testing. A stratified sampling strategy was applied to preserve the original class distribution of default and non-default cases across both subsets.

4. Methodology

This study adopts an experimental methodology to evaluate the effectiveness of ensemble learning and data preprocessing strategies for loan default prediction. A strong gradient-boosted decision tree model, LightGBM (LGBM), was used as the primary baseline and compared against a stacked ensemble model. Multiple preprocessing and modeling techniques were systematically evaluated and embedded within the training pipeline to assess whether they yielded measurable improvements in predictive performance.

4.1 Baseline Model: LightGBM

LightGBM was selected as the baseline model due to its strong performance on tabular financial data, ability to handle missing values natively, and computational efficiency on large datasets. The baseline LGBM model was trained using the aggregated and engineered feature set described in the preprocessing section. Default hyperparameters were initially used, followed by controlled tuning to establish a competitive benchmark.

4.2 Stacked Ensemble Model

To assess whether combining heterogeneous learners could improve predictive performance, a stacked ensemble model was constructed. The ensemble consists of the following base learners:

- LightGBM
- Random Forest Regressor
- Elastic Net Regressor

These base models were trained independently on the same feature set. Their out-of-fold predictions were then used as inputs to a **Logistic Regression meta-model**, which learned how to optimally combine the base learner outputs. The class weights used for the Logistic Regression is **1:1.25**, favoring class 1 predictions, to emphasize the importance of Recall i.e. the ability to flag potential defaulters. This stacking strategy was chosen to capture complementary strengths across tree-based, linear, and regularized models while maintaining interpretability at the meta-learning stage.

4.3 Evaluation of Preprocessing and Data Handling Strategies

Several preprocessing and data-handling techniques were evaluated to determine their impact on model performance. Each technique was embedded within the modeling pipeline and assessed independently and in combination with both the LGBM baseline and the stacked ensemble.

4.3.1 Class Imbalance Handling

- **Random undersampling** was applied to the majority (non-default) class to address severe class imbalance. Performance with and without undersampling was compared to evaluate its effect on the model's ability to identify default cases.

4.3.2 Feature Transformation and Dimensionality Reduction

- **Principal Component Analysis (PCA)** was evaluated to determine whether dimensionality reduction could improve generalization by reducing feature redundancy and noise.
- **Feature selection using LightGBM feature importance** was applied to retain the most informative predictors and remove weak or noisy features.

4.3.3 Feature Scaling

- **Standard scaling** was tested to assess its impact on models sensitive to feature magnitude, particularly Elastic Net and Logistic Regression. The effect of scaling on overall ensemble performance was evaluated.

4.4 Missing Value Handling Strategies

Given the high degree of missingness in the dataset, multiple imputation strategies were explored and compared:

- **Zero imputation (Missing: 0):** Missing values were replaced with zeros to test whether explicit absence signals were informative.
- **Mean imputation (Missing: mean):** Numerical features were imputed using feature-wise means.
- **k-Nearest Neighbors imputation (Missing: kNN, k=5):** Missing values were imputed based on the average of the five nearest neighbors in feature space.
- **Feature-as-target modeling (EXT_SOURCE_1):**
 - A predictive model was trained with *EXT_SOURCE_1* treated as the target variable, using other features as inputs.
 - To identify the most suitable predictive model for this task, multiple regression algorithms were evaluated: Random Forest Regressor (RFR), Support Vector Regressor (SVR), LightGBM Regressor (LGBMR)
 - The resulting predictions were evaluated in two ways: (1) adding the predicted values as an additional feature, and (2) imputing missing values of *EXT_SOURCE_1* directly.

These strategies were compared to determine whether more complex imputation methods provided tangible benefits over simpler approaches.

4.5 Hyperparameter Optimization

To ensure fair model comparison, **Optuna** was employed for automated hyperparameter optimization. Search spaces were defined for both LightGBM and selected components of the stacked ensemble. Optimization was conducted using cross-validated performance metrics, and the best-performing configurations were retained for final evaluation.

4.6 Model Comparison and Selection

All models and preprocessing configurations were evaluated using the same stratified train–test split and performance metrics. Improvements were assessed relative to the baseline LightGBM model. Only preprocessing and modeling techniques that demonstrated consistent and meaningful performance gains were retained in the final pipeline.

This methodology ensures that model complexity is justified by empirical gains and that the final model reflects a balance between predictive performance, robustness, and practical applicability in real-world credit risk settings.

4.7 Evaluation Metrics

4.7.1 Loan Default Prediction

For the primary task of loan default prediction, **F1-score** was used as the main metric for model selection. The F1-score provides a balanced measure of precision and recall, which is particularly important in this highly imbalanced classification setting where both false positives (misclassifying a non-default as default) and false negatives (misclassifying a default as non-default) carry significant consequences.

As a secondary metric, **Recall** was monitored to ensure that models maintained sensitivity toward detecting default cases. Maximizing recall helps minimize the risk of overlooking high-risk borrowers, aligning model evaluation with practical credit risk priorities.

4.7.2 EXT_SOURCE_1 Feature-as-Target Prediction

For the feature-as-target regression task on *EXT_SOURCE_1*, multiple regression metrics were employed to evaluate the predictive quality of different imputation models:

- **R² Score (Coefficient of Determination):** Measures the proportion of variance in *EXT_SOURCE_1* explained by the model, providing a normalized assessment of fit.
- **Mean Absolute Error (MAE):** Captures the average absolute difference between predicted and observed values, offering an interpretable measure of prediction error in the original scale.
- **Mean Squared Error (MSE):** Emphasizes larger errors due to squaring, highlighting the model’s ability to avoid extreme mispredictions that may adversely impact downstream credit risk prediction.

By evaluating candidate models using these metrics, the best-performing algorithm for *EXT_SOURCE_1* imputation was selected before integration into the final default prediction pipeline.

5. Results and Discussion

The predictive performance of the baseline LightGBM model was compared against a stacked ensemble model under multiple preprocessing and data handling strategies. Table 1 and table 2 summarizes the performance metrics, including **ROC AUC, Accuracy, Precision, Recall, and F1-score**, for LightGBM and the stacked model, respectively. Table 3 shows the performance of the models when predicting *EXT_SOURCE_1* as the target variable.

Table 1 (LightGBM Performance)

	ROC AUC	Accuracy	Precision	Recall	F1 Score
Baseline	0.70975	0.76092	0.19906	0.64874	0.30464

Undersampling	0.70685	0.70685	0.70515	0.71098	0.70805
PCA	0.69436	0.69436	0.69346	0.69668	0.69507
LGBM Top features	0.70896	0.70896	0.70829	0.71057	0.70943
Standard Scaling	0.70755	0.70755	0.70806	0.70634	0.70720
Missing: 0	0.70775	0.70775	0.70834	0.70634	0.70734
Missing: kNN k=5	0.70735	0.70735	0.70644	0.70957	0.70800
Missing: mean	0.70725	0.70725	0.70654	0.70896	0.70775
Missing: feature-as-target EXT_SOURCE_1 (add column)	0.70524	0.70524	0.70331	0.70997	0.70663
Missing: feature-as-target EXT_SOURCE_1 (impute null)	0.70765	0.70524	0.70695	0.70937	0.70815
Optuna	0.71037	0.71037	0.71003	0.71118	0.71061

Table 2 (Stacked Model Performance)

	ROC AUC	Accuracy	Precision	Recall	F1 Score
Baseline	0.50919	0.91945	0.52941	0.01994	0.03843
Undersampling	0.70544	0.70544	0.68149	0.77140	0.72367
PCA	0.69476	0.69476	0.66876	0.77180	0.71660
LGBM Top features	0.70745	0.70745	0.68354	0.77261	0.72535
Standard Scaling	0.70604	0.70604	0.68222	0.77140	0.72408
Missing: 0	0.70554	0.70554	0.68331	0.76616	0.72237
Missing: kNN k=5	0.70514	0.70514	0.68158	0.76999	0.72309
Missing: mean	0.70574	0.70574	0.68166	0.77200	0.72403
Missing: feature-as-target EXT_SOURCE_1 (add column)	0.70594	0.70594	0.68243	0.77039	0.72375
Missing: feature-as-target EXT_SOURCE_1 (impute null)	0.70524	0.70524	0.68196	0.76918	0.72295
Optuna	0.70554	0.70554	0.68338	0.76596	0.72232

Table 3 (EXT_SOURCE_1)

	R2	MAE	MSE
RFR	0.524226	0.121091	0.02198
SVR	-33.5286	1.2208	1.5948
LGBMR	0.99239	0.01373	0.00035

5.1 Impact of Class Imbalance Handling

Applying **random undersampling** led to a substantial improvement for both models, particularly in F1-score and Recall. For the stacked ensemble, the F1-score increased from 0.0384 to **0.7237**, and Recall reached **0.7714**. LightGBM also benefited, with the F1-score increasing from **0.3046** to **0.7081**. These results confirm that mitigating class imbalance is critical in improving predictive performance for highly skewed credit datasets.

5.2 Feature Selection and Dimensionality Reduction

Feature selection based on **LightGBM top features** further improved performance for both models. The stacked model achieved the highest F1-score of **0.7254** and a Recall of **0.7726**, highlighting the importance of retaining only the most informative features while removing noise. In contrast, **PCA** did not improve performance, suggesting that unsupervised dimensionality reduction may discard critical predictive information in this domain.

5.3 Scaling and Encoding

Applying **standard scaling** had minimal effect, as tree-based models like LightGBM are largely insensitive to feature magnitudes. Similarly, encoding strategies were handled consistently across experiments, and no significant differences were observed in performance metrics due to scaling.

5.4 Missing Value Imputation Strategies

Various missing value handling strategies were evaluated, including zero imputation, mean imputation, kNN imputation, and feature-as-target modeling for *EXT_SOURCE_1*. Across both models, these strategies yielded minor variations in F1-score (within ~0.005 for LightGBM and ~0.003 for the stacked model). Feature-as-target imputation (both adding a predicted column and imputing missing values) performed comparably to simpler methods, suggesting that complex imputation provided limited additional predictive benefit once key features were selected and class imbalance was addressed. This may also indicate that the missing values themselves carry informative signals for the model.

5.5 Hyperparameter Optimization

Automated hyperparameter tuning with **Optuna** provided modest gains for the LightGBM baseline (F1-score increased to **0.7106**) but did not yield significant improvements for the stacked ensemble. This indicates that preprocessing and feature selection had a larger impact on model performance than fine-tuning hyperparameters for this dataset.

5.6 Overall Model Comparison

The results demonstrate that the **stacked ensemble outperforms the LightGBM** when preprocessing is carefully applied, particularly using **random undersampling** and **LightGBM-based feature selection**. These two interventions were the most influential in improving F1-score and Recall, confirming their

critical role in credit risk prediction for highly imbalanced datasets. Other strategies, such as PCA, scaling, and complex missing value imputation, contributed minimally and were largely redundant once these key preprocessing steps were applied.

The findings highlight three important insights for credit risk modeling:

1. Class imbalance handling is essential for detecting default cases effectively.
2. Targeted feature selection leveraging domain-informed importance improves model generalization by reducing noise.
3. Complex imputation or dimensionality reduction may offer limited benefits when the dataset is large, noisy, and already enriched with engineered features.
4. Stacking tree-based models with regression models improves model performance

Overall, the stacked ensemble coupled with random undersampling and LightGBM top feature selection achieved the best balance of precision, recall, and F1-score, making it the recommended pipeline for robust loan default prediction in this context.

6. Conclusion

The results highlight the importance of **combining domain knowledge, targeted feature engineering, and ensemble learning** to handle large, noisy, and imbalanced financial datasets effectively. The stacked ensemble consistently outperformed the baseline LightGBM model when these preprocessing strategies were applied, achieving a robust balance between sensitivity to defaults and overall predictive accuracy.

In practice, this study demonstrates that careful preprocessing and domain-informed feature selection can yield substantial performance improvements without relying solely on model complexity or hyperparameter tuning. Models can also be customized for real-world deployment by adjusting class weights or the precision–recall trade-off to meet the specific operational objectives of a lending institution. This flexibility allows practitioners to prioritize minimizing defaults, maximizing approved loans, or balancing both considerations depending on the financial context.

These findings provide actionable insights for practitioners aiming to develop reliable, interpretable, and high-performing credit risk models in real-world lending scenarios.

7. Recommendations

Based on the findings of this study, several directions are suggested to further improve model performance and practical applicability in credit risk prediction:

1. Expand Model Stacking Experiments

Further experimentation with alternative stacking strategies and diverse base learners may uncover combinations that yield additional predictive gains. Exploring heterogeneous ensembles and different meta-model configurations could enhance the robustness and generalization of the

final model.

2. **Develop Additional Features**

Generating new features, particularly those informed by domain knowledge, may improve the model's ability to capture subtle borrower behaviors and credit patterns. Incorporating external data sources or more granular behavioral metrics could provide additional predictive signals for default risk.

3. **Incorporate Counterfactual Explanations and Recourse**

Implementing tools such as CARLA or DiCE to generate counterfactuals for predicted default cases (class 1) can improve model transparency and provide actionable insights for both lenders and borrowers. Counterfactual analysis can help identify which feature changes could reduce default risk, enabling responsible lending decisions and enhancing the practical utility of the predictive model.

By pursuing these recommendations, future work can not only improve predictive performance but also enhance interpretability, fairness, and actionable value in real-world credit risk applications.

8. References

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